

```
def main():
    es = Elasticsearch(hosts='h-mstr')
    for path,name,text in hdfs_input():
        doc = {'path':path,'title':name, 'text':text}
        es.index(index='pdfdocs', doc_type='text', body= doc)
if __name__ == "__main__" :
    main()
```

Run the code in the following command on *h-mstr*:

```
$ hadoop jar /usr/share/java/hadoop/hadoop-streaming.jar \
-files indexing_mapper.py -mapper indexing_mapper.py \
-input document_files.txt -output es.out
```

The following URLs will give you more information about the allocation and status of the index.

```
http://h-mstr:9200/_cat/allocation
http://h-mstr:9200/_cat/indices
```

## Using a Pig script to create an index

The Fedora 20 repositories do not as yet have the Pig distribution. It will be included in Fedora 21. So, download and install Pig from the Apache site <http://pig.apache.org/releases.html> on each of the virtual machines.

You will also need to install the Elasticsearch-Hadoop plugin on these systems. For example, you may run the following commands from *h-mstr*:

```
$ sudo /usr/share/elasticsearch/bin/plugin \
-u http://download.elasticsearch.org/hadoop/elasticsearch-hadoop-2.1.0.Beta2.zip \
-i elasticsearch-hadoop

$ ssh -t fedora@h-slv1 sudo /usr/share/elasticsearch/bin/plugin \
-u http://download.elasticsearch.org/hadoop/elasticsearch-hadoop-2.1.0.Beta2.zip \
-i elasticsearch-hadoop

$ ssh -t fedora@h-slv2 sudo /usr/share/elasticsearch/bin/plugin \
-u http://download.elasticsearch.org/hadoop/elasticsearch-hadoop-2.1.0.Beta2.zip \
-i elasticsearch-hadoop
```

The Pig script, *indexing.pig*, for creating the index is just four lines. The *elasticsearch-hadoop jar* file has to be registered. The Hadoop text file is loaded. The tuple (path, text, title) is stored in (indexed by) Elasticsearch storage.

```
REGISTER /usr/share/elasticsearch/plugins/hadoop/dist/
elasticsearch-hadoop-2.1.0.Beta2.jar;
A = LOAD 'document_files.txt' USING PigStorage() AS
(path:chararray, title:chararray, text:chararray);
B = FOREACH A GENERATE path, text, title ;
STORE B INTO 'docs/text' USING org.elasticsearch.hadoop.pig.
EsStorage();
```

You can check the status of the indices and compare the *pdfdocs* index created earlier with the *docs* index created by running the Pig script:

```
[fedora@h-mstr ~]$ pig indexing.pig
```

The ultimate test is to compare the results of the two indices; e.g., you can browse the Elasticsearch index searching for 'Python' in the contents and displaying up to 25 values. Only the values of the fields path and title will be displayed.

```
http://h-mstr:9200/docs/_search?pretty=true&size=25&fields=path,title&q=text:python
```

The more flexible option is to use a *json* string to query as follows (for details, go to <http://www.elasticsearch.org/guide/en/elasticsearch/reference/current/query-dsl.html>):

```
curl "h-mstr:9200/pdfdocs/_search?pretty=true" -d '{
  "fields": ["path", "title"],
  "size": 25,
  "query": { "query_string": { "query": "python" } }'}
```

If all has gone well, you should get the same answers for the queries—whether you use the *docs* or *pdfdocs* indices. 

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